

Wang Proposition 5.2

Mu-Tao Wang, <i>Mean Curvature Flow of Surfaces in Einstein Four-Manifolds</i> , Section 5,			
Proposition 5.2	323--324	Gaussian density	1
		varifold	“
” “multiplicity one”		Gaussian density	2
density	1	White	

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Danus Wang Huisken White

Type-I $\implies$ $\implies$ $\implies \Theta(y_0, t_0) = 1 \implies$.

Type-I	$ A^\lambda $		
		Huisken	
	multiplicity one		sheets
	Gaussian density 1	m	density m
density 1		White	

Proposition 5.2

multiplicity one Kähler Huisken

1.

Wang Proposition 5.2

$$F : \Sigma \times [0, t_0) \longrightarrow M^4 \hookrightarrow \mathbb{R}^N,$$

$$\eta_t = *F_t^* \omega > \delta > 0, \qquad |A_t|^2 \leq \frac{C}{t_0 - t}.$$
(1.1)

$$F \qquad \qquad t_0$$

	immersions	Proposition 5.2		embeddedness
Theorem D	embedded	“	”	

$\lambda_i \rightarrow \infty$	$s_i \rightarrow -1$	$\Sigma_{s_i}^{\lambda_i}$	Huisken [3]
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$$\Sigma_{s_i}^{\lambda_i} \longrightarrow \Sigma_{-1}^\infty$$
(1.2)

$F^\perp = 0$	multiplicity-one plane	Gaussian integral	1
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(1.1)--(1.2)

2.

(y_0, t_0)

$$F^\lambda(x, s) = \lambda \big(F(x, t_0 + \lambda^{-2}s) - y_0 \big), \qquad s < 0.$$
(2.1)

Σ_s^λ	λ	λ^{-1}
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$$|A^\lambda|^2(x, s) = \lambda^{-2} |A|^2 \left(x, t_0 + \frac{s}{\lambda^2} \right).$$
(2.2)

(1.1) $t_0 + s/\lambda^2 \in [0, t_0)$

$$\begin{aligned} |A^\lambda|^2(x, s) &\leq \lambda^{-2} \frac{C}{t_0 - (t_0 + s/\lambda^2)} \\ &= \frac{C}{-s}. \end{aligned}$$
(2.3)

$$[a,b]\Subset (-\infty,0),$$

$$\lambda$$

$$\sup_{s\in[a,b]}|A^\lambda|^2\leq \frac{C}{-b}.$$
(2.4)

$$t\uparrow t_0$$

$$|A|^2(x,t)=\lambda^2|A^\lambda|^2(x,s)$$

$$(t_0-t)^{-1}$$
“
”

$$3. \qquad \qquad \qquad \text{“} \qquad \qquad \qquad \text{”}$$

$$(2.4)$$

$$|\nabla^k A^{\lambda_i}|\leq C_{k,K}$$
(3.1)

$$C^\infty$$

$$\text{Huisken 1990} \qquad \text{Proposition 2.3} \qquad \qquad \qquad \text{Proposition 3.4}$$

$$\text{Brakke}$$

$$1.$$

$$2. \qquad \qquad \qquad m$$

$$3. \qquad \text{degree-one} \qquad \qquad \qquad \text{sheet}$$

$$\text{integral varifold} \qquad \qquad m \qquad \qquad \qquad \text{multiplicity one}$$

$$\text{sheet}$$

$$4.$$

$$\text{Wang} \qquad \qquad \qquad s_i \rightarrow -1$$

$$Q:=(h_{31k}-h_{42k})^2+(h_{32k}+h_{41k})^2=0$$
(4.1)

$$(4.1)$$

$$H=0,\qquad \nabla \eta=0.$$
(4.2)

$$\text{Huisken} \qquad \qquad \qquad \text{shrinker}$$

$$H+\frac{1}{2}F^\perp=0. \tag{4.3}$$

(4.2)--(4.3)

$$F^\perp=0. \tag{4.4}$$

(4.4)

$$P$$

$$\operatorname{spt} V_{-1}=P \tag{4.5}$$

integral varifold

$$V_{-1}=m|P|,\qquad m\in\mathbb{N}, \tag{4.6}$$

(4.1)--(4.5) \hspace{1cm} m

$$m\geq 2\qquad \text{varifold }m|P|$$

$$A=H=F^\perp=\nabla\eta=Q=0.$$

5.

5.1

$$P=\{z=0\}\subset\mathbb{R}^3$$

$$\varepsilon_i\downarrow 0,\qquad R_i\uparrow\infty.$$

$$\Gamma_i$$

$$r\leq R_i$$

$$z=\varepsilon_i,\qquad z=-\varepsilon_i. \tag{5.1}$$

$$R_i\leq r\leq R_i+L$$

$$z=1\qquad z=-1$$

$$R_i+L$$

$$k\qquad i\qquad C_k$$

$$\sup_{\Gamma_i}|\nabla^kA_{\Gamma_i}|\leq C_k. \tag{5.2}$$

$$\Gamma_i$$

$$S^2$$

$$K\subset\mathbb{R}^3\qquad i\qquad K$$

(5.1)

$$\Phi\qquad \text{varifold}$$

$$i$$

$$\Phi$$

$$\begin{aligned} &\int_{\Gamma_i} \Phi(x, T_x \Gamma_i) \, d\mathcal{H}^2(x) \\ &= \int_P \Phi(y + \varepsilon_i \nu, P) \, d\mathcal{H}^2(y) + \int_P \Phi(y - \varepsilon_i \nu, P) \, d\mathcal{H}^2(y), \end{aligned} \tag{5.3}$$

$$\begin{aligned} &\Phi \\ &|\Gamma_i| \longrightarrow 2|P| \quad \ll \text{varifold} \gg. \end{aligned} \tag{5.4}$$

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-
-
-

sheets
 i
 i
 sheet separation

5.2

Γ_i

Wang	multiplicity one
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Proposition 5.2
 symplectic

$$P \subset \mathbb{C}^2 \qquad \eta = 1 \qquad m|P|$$

Kähler
 m

6. Gaussian density

$$s=-1$$

$$G(x)=\frac{1}{4\pi}e^{-|x|^2/4}.\tag{6.1}$$

P

$$\begin{aligned} \int_P G \, d\mathcal{H}^2 &= \frac{1}{4\pi} \int_0^{2\pi} \int_0^\infty e^{-r^2/4} r \, dr \, d\theta \\ &= 1. \end{aligned} \tag{6.2}$$

$$V_{-1}=m|P|$$

$$\Theta(y_0,t_0)=\int G\,d\|V_{-1}\|=m\int_P G\,d\mathcal{H}^2=m. \tag{6.3}$$

$$\mathbb{I}\mathbb{I}\mathbb{I}\mathbb{I}+\mathbb{I}\mathbb{I}\,m\Longrightarrow\Theta=m, \tag{6.4}$$

$$“\hspace{1.5cm}\Theta=1”$$

$$m\geq 2\hspace{1cm}\begin{array}{lll} \text{White}&&\text{White}\hspace{1cm}\text{Gaussian density ratio} \end{array}\hspace{1cm}1 \tag{6.3}$$

7. density 2

7.1

$$F:\Sigma\times[0,t_0)\longrightarrow M^4$$

Kähler

$$*F_t^*\omega\geq\delta>0,\qquad |A_t|^2\leq\frac{C}{t_0-t}. \tag{7.1}$$

Wangtangent flow

Gaussian density

$$\Theta(y_0,t_0)<2. \tag{7.2}$$

$$\varepsilon>0$$

$$[0,t_0+\varepsilon).$$

$$\begin{array}{llll} (y_0,t_0) & (7.1) & & \text{tangent flow} \\ -1 & P & \text{varifold} & \end{array}$$

$$V_{-1}=m|P|,\qquad m\in\mathbb{N}. \tag{7.3}$$

$$\text{Gaussian} \hspace{10cm} (6.3)$$

$$\Theta(y_0,t_0)=m. \tag{7.4}$$

$$(7.2) \hspace{1cm} m \hspace{1cm} m<2$$

$$m=1,\qquad \Theta(y_0,t_0)=1. \tag{7.5}$$

$$\varepsilon_{\mathrm{W}}>0\hspace{1cm}\text{White}$$

$$1 < 1 + \varepsilon_W,$$

$$\text{White} \qquad (y_0,t_0)$$

$$\Sigma \qquad \qquad \qquad t_0$$

$$\sup_{\Sigma\times [t_0-\tau,t_0)}|A|<\infty \tag{7.6}$$

$$\tau>0$$

$$\sup_{\Sigma\times [0,t_0)}|A|<\infty.$$

$$\varepsilon>0\qquad\qquad\qquad [0,t_0+\varepsilon)$$

$$7.2$$

$$(7.2) \qquad m \geq 2$$

1. tangent flow
- unit-weight plane
2.
- degree-one
3.
- varifold
- $|P|$
4.
- Kähler

$$\text{Bamler--Kleiner} \qquad \mathbb{R}^3$$

$$8. \quad \text{“} \qquad \qquad \text{”}$$

$$K\Subset P$$

$$X_i(p)=p+u_i(p),\qquad \|u_i\|_{C^k(K)}\longrightarrow 0\quad \text{as}\quad k.\tag{8.1}$$

$$\text{Jacobian } J_i \qquad 1 \qquad \qquad \phi$$

$$\int_{\Sigma_i}\phi\,d\mathcal{H}^2=\int_P\phi(X_i(p))J_i(p)\,d\mathcal{H}^2(p)\longrightarrow\int_P\phi(p)\,d\mathcal{H}^2(p).\tag{8.2}$$

$$\text{Gaussian}$$

$$\int_{\Sigma_i}G\,d\mathcal{H}^2\longrightarrow\int_PG\,d\mathcal{H}^2=1.\tag{8.3}$$

$$\text{“} \qquad \qquad \text{density 1”} \qquad \qquad \textbf{degree-one} \qquad \qquad \text{Gaussian}$$

$$\text{“} \qquad \qquad \qquad \qquad \qquad \qquad \qquad \text{”}$$

9.

(y_0, t_0)

$$D_\lambda(y, t) = (\lambda(y - y_0), \lambda^2(t - t_0))$$

(9.1)

- (y_0, t_0) multiplicity-one density 1
- density 0
- shrinker

Wang

y_0

density

blow-up

White

10.

weighted Gaussian density

7

Gaussian density

$$\Theta^w = \int w \, G \, d\|V\|,$$

- $w \geq 1$ $\cos^{-p} \alpha$ $m \geq 2$ $\Theta^w \geq 2$ $\Theta^w < 2$
tangent density
- $w = 1 - \eta$ $w = 0$ m “weighted density 2”
 $m \geq 2$
weighted density Wang Gaussian
density

11.

- “Type-I ”
- “ ”
- “ ”
- “ ”
- “ Gaussian integral 1 White ”

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- [1] M.-T. Wang, *Mean Curvature Flow of Surfaces in Einstein Four-Manifolds*, Journal of Differential Geometry 57 (2001), 301--338. DOI: 10.4310/jdg/1090348113; arXiv: math/0110019.
- [2] G. Huisken, *Asymptotic Behavior for Singularities of the Mean Curvature Flow*, Journal of Differential Geometry 31 (1990), 285--299. DOI: 10.4310/jdg/1214444099.
- [3] B. White, *A Local Regularity Theorem for Mean Curvature Flow*, Annals of Mathematics 161 (2005), 1487--1519. DOI: 10.4007/annals.2005.161.1487.
- [4] H. Li and B. Wang, *The Extension Problem of the Mean Curvature Flow (I)*, arXiv:1608.02832.
- [5] R. H. Bamler and B. Kleiner, *On the Multiplicity One Conjecture for Mean Curvature Flows of Surfaces*, arXiv:2312.02106.